

MIDDLE ROCK LIGHT, AK, USA

WMO: 994670

Lat: 61.0811N

Lon: 146.6530W

Elev: 0

StdP: 14.70

Time Zone: -9.00 (NAK)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 11.7	(c) 14.7	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 40.0	(k) 22.0	(l) 35.9	(m) 22.3	(n) 20.5	(o) 50	(p) 0.778

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.6	(c) 63.5	(d) N/A	(e) 60.9	(f) N/A	(g) 59.2	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 15.2	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
33.7	27.0	22.9	8.8	70.9	3.2	6.1	6.5	75.3	4.6	78.8	2.8	82.2	0.5	86.7	
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.3	28.1	29.5	31.4	38.2	46.2	51.3	53.6	52.6	48.0	41.3	32.4	30.4
	DBStd	10.55	7.30	5.67	5.51	4.42	4.25	3.44	3.05	3.24	3.00	3.82	6.50	5.67
	HDD50	3831	678	574	578	355	133	22	5	7	73	269	528	609
	HDD65	9016	1143	994	1043	805	583	413	353	385	511	734	978	1074
	CDD50	290	0	0	0	1	15	59	116	87	12	0	0	0
	CDD65	1	0	0	0	0	0	1	0	0	0	0	0	0
	CDH74	6	0	0	0	0	0	3	1	2	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	10.5	13.4	12.1	12.6	8.4	9.4	9.7	9.4	8.9	8.2	9.0	13.0	12.4
Precipitation	PrecAvg	44.3	4.0	3.7	2.7	2.2	2.0	2.2	3.3	4.6	6.0	5.0	3.9	4.4
	PrecMax	55.2	10.4	9.3	5.8	4.4	4.4	3.9	5.2	11.1	10.2	12.6	9.4	10.0
	PrecMin	30.4	0.4	0.0	0.6	0.5	0.5	0.7	1.7	0.6	2.7	2.2	0.1	1.0
	PrecStd	6.4	2.3	2.1	1.7	1.1	1.1	0.8	1.0	2.5	2.1	2.7	2.6	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.4	40.4	44.2	54.7	62.3	67.5	68.0	69.2	59.4	52.0	45.8	41.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	39.7	38.4	41.6	49.3	58.1	61.6	62.6	63.2	56.3	49.4	42.8	39.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	37.8	37.2	39.8	46.1	55.0	58.5	60.5	60.4	54.4	47.7	41.2	38.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	36.2	36.0	38.1	43.8	52.4	56.4	59.2	58.3	52.5	46.2	39.8	36.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.0	5.7	6.3	6.6	6.5	7.6	7.9	6.7	5.2	5.0	5.0
		MCDBR	6.5	6.1	7.2	9.5	11.3	10.9	10.9	11.9	9.7	7.5	5.7	6.3
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.267	0.328	0.333	0.388	0.411	0.401	0.410	0.405	0.348	0.326	0.282	0.263
	taud		2.002	2.012	2.049	1.951	2.019	2.182	2.214	2.218	2.338	2.179	1.983	1.901
	Ebn at Noon		134	192	241	249	254	261	255	245	241	195	120	75
	Edh at Noon		14	26	35	47	48	42	39	36	27	22	14	10
All-Sky Solar Radiation	RadAvg		89	307	799	1364	1720	1793	1600	1217	785	359	128	48
	RadStd		10	44	82	111	130	100	103	118	104	40	22	5

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)		N/A	+2.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page